

A. 지역/연령별 가계자금 금액 및 금융자산 보유의 특징 이해를 위한 다변량
자료 분석

1. Data

[작업경로 설정 및 데이터 셋팅]

```
setwd("C:/Users/sec/Desktop/family/data")
X = read.csv("family.csv",header=T)
colnames(X)<-c("utility bills", "livingfund", "tax"
              , "annualpension", "savings", "loan")
rownames(X)<-c(101:117, 201:217, 301:317, 401:417,
              501:517, 601:617, 701:717)
```

[분석을 위한 기초적 데이터 변환]

```
X
class(X)
X<-as.matrix(X)
n<-nrow(X)
xbar<-t(X)%*%matrix(1,n,1)/n
I<-diag(n)
J<-matrix(1,n,n)
H<-I-1/n*J
Y<-H%*%X
S<-t(Y)%*%Y/(n-1)
D<-diag(1/sqrt(diag(S)))
Z<-H%*%X%*%D
colnames(Z)<-colnames(X)
R<-t(Z)%*%Z/(n-1)
[표 1.1.]
summary(X)
```

2. Data Description

[그림 2.1.]

```
plot(X)
```

[표 2.1.]

```
cor(X)
```

[표 2.2., 그림 2.2.]

```
library("MVN") # for mardia test
result = mvn(X,
              mvnTest = "mardia",
              multivariatePlot = "qq")
result
```

[표 2.3., 그림 2.3.]

```
library("MVT")
rownames<-rownames(X)
R=round(cor(X),3)
R
eigen.R=eigen(R)
round(eigen.R$values, 2) # Eigenvalues
V=round(eigen.R$vectors, 2) # Eigenvectors
V6=V[,1:6]
V6
Z=scale(X, scale=T) # Standardized Data Matrix
Z
P=Z%*%V6 # PCs Scores
round(P, 3)
round(Z, 3)
result = mvn(P,
              mvnTest = "mardia",
              multivariatePlot = "qq")
result
```

3. Principal Component Analysis(PCA)

[표 3.1.]

```
cov(X)
```

[표 3.2.]

```
gof=eigen.R$values/sum(eigen.R$values)*100 # Goodness-of fit  
round(gof, 2)
```

[그림 3.1.]

```
plot(eigen.R$values, type="b",  
      main="Scree Graph",  
      xlab="Component Number",  
      ylab="Eigenvalue")
```

[표 3.3.]

```
V3=V[,1:3]  
V3
```

[그림 3.2.]

```
plot(P[,1], P[, 2],  
      main="Plot of PCs Scores", xlab="1st PC", ylab="2nd PC")  
text(P[,1], P[, 2], labels=rownames, cex=0.8, col="blue", pos=3)  
abline(v=0, h=0)
```

[그림 3.3.]

```
plot(P[,1], P[, 3],  
      main="Plot of PCs Scores", xlab="1st PC", ylab="3rd PC")  
text(P[,1], P[, 3], labels=rownames, cex=0.8, col="blue", pos=3)  
abline(v=0, h=0)
```

[그림 3.4.]

```
plot(P[,2], P[, 3],  
      main="Plot of PCs Scores", xlab="2nd PC", ylab="3rd PC")  
text(P[,2], P[, 3], labels=rownames, cex=0.8, col="blue", pos=3)  
abline(v=0, h=0)
```

[그림 3.5.]

```
pcasvd.Z<-prcomp(X, scale=T)
summary(pcasvd.Z)
round(pcasvd.Z$rotation, 3)
pcasvd.Z$scale
biplot(pcasvd.Z, scale=0, xlab="1st PC",ylab="2nd PC",
       main="PC Biplot")
abline(v=0, h=0)
```

4. Factor Analysis (FA)

[FA를 위한 기본 설정]

```
p=ncol(X)
eigen.R=eigen(R)
round(eigen.R$values, 2) # Eigenvalues
V=round(eigen.R$vectors, 2) # Eigenvectors
V
```

[표 4.1.]

```
gof=eigen.R$values/p*100 # Goodness-of fit
round(gof, 3) # contribution rate
```

[그림 4.1.]

```
plot(eigen.R$values, type="b", main="Scree Graph",
     xlab="Factor Number", ylab="Eigenvalue")
```

[그림 4.2.]

```
pcfa<-principal(Z, nfactors=3, rotate="none")
pcfa
lim<-range(pretty(L))
plot(L[,1], L[,2],main="PC Factor Loadings : f1 and f2", xlab="f1", ylab="f2",
     xlim=lim, ylim=lim)
text(L[,1], L[, 2], labels=rownames(L), cex=0.8, col="blue", pos=1)
abline(v=0, h=0)
arrows(0,0, L[,1], L[, 2], col=2, code=2, length=0.1)
plot(L[,1], L[,3],main="PC Factor Loadings : f1 and f3", xlab="f1", ylab="f3",
     xlim=lim, ylim=lim)
```

```

text(L[,1], L[, 3], labels=rownames(L), cex=0.8, col="blue", pos=1)
abline(v=0, h=0)
arrows(0,0, L[,1], L[, 3], col=2, code=2, length=0.1)
plot(L[,2], L[,3],main="PC Factor Loadings : f2 and f3", xlab="f2", ylab="f3",
      xlim=lim, ylim=lim)
text(L[,2], L[, 3], labels=rownames(L), cex=0.8, col="blue", pos=1)
abline(v=0, h=0)
arrows(0,0, L[,2], L[, 3], col=2, code=2, length=0.1)

```

[그림 4.3.]

```

fpc=pcfa$scores
round(fpc, 3)
par(pty="s")
lim<-range(pretty(fpc))
plot(fpc[,1], fpc[,2],main="Factor Scores : f1 and f2", xlab="f1", ylab="f2",
      xlim=lim, ylim=lim)
text(fpc[,1], fpc[,2], labels=rownames(fpc), cex=0.6, col="blue")
abline(v=0, h=0)
plot(fpc[,1], fpc[,3],main="Factor Scores : f1 and f3", xlab="f1", ylab="f3",
      xlim=lim, ylim=lim)
text(fpc[,1], fpc[,3], labels=rownames(fpc), cex=0.6, col="blue")
abline(v=0, h=0)
plot(fpc[,2], fpc[,3],main="Factor Scores : f2 and f3", xlab="f2", ylab="f3",
      xlim=lim, ylim=lim)
text(fpc[,2], fpc[,3], labels=rownames(fpc), cex=0.6, col="blue")
abline(v=0, h=0)

```

[그림 4.4.]

```

svd.Z <- svd(Z)
U <- svd.Z$u
V <- svd.Z$v
D <- diag(svd.Z$d)
F <- (sqrt(n-1)*U)[,1:2] # Factor Scores Matrix : F
L <- (sqrt(1/(n-1))*V%*%D)[,1:2] # Factor Loadings Matrix : Lambda
C<- rbind(F, L)
rownames(F)<-rownames(X)
rownames(L)<-colnames(X)

```

```

varimax<-varimax(L)
Lt = varimax$loadings
T=varimax$rotmat
T
Ft= F%*%T
biplot(Ft,Lt, xlab="f1",ylab="f2", main="Varimax Rotated Biplot",
        xlim=lim2,ylim=lim2,cex=0.5,pch=16)
abline(v=0,h=0)

```

```

[¶ 4.2., ¶ 4.3.]
rownames<-rownames(X)
R=round(cor(X),3)
R
eigen.R=eigen(R)
round(eigen.R$values, 2)
V=round(eigen.R$vectors, 2)
V6=V[,1:6]
V6
Z=scale(X, scale=T)
Z
P=Z%*%V6
round(P, 6)
mlfa<-factanal(P, factors = 3, rotation="varimax" )
mlfa

```

```

[¶ 4.4]
L=pcfa$loading[, 1:3]
round(L, 3)
Psi=pcfa$uniquenesses
Psi
Rm = R-(L%*%t(L) + diag(Psi))
round(Rm, 3)
###
L=mlfa$loading[,1:2]
Psi=mlfa$uniquenesses
Rm = cor(P)-(L%*%t(L) + diag(Psi))
round(Rm, 3)

```

5. Cluster Analysis (CA)

[그림 5.1.]

```
n<-nrow(X)
xbar<-t(X)%*%matrix(1,n,1)/n
I<-diag(n)
J<-matrix(1,n,n)
H<-I-1/n*J
Y<-H%*%as.matrix(X)
S<-t(Y)%*%Y/(n-1)
D<-diag(1/sqrt(diag(S)))
Z<-Y%*%D
colnames(Z)<-colnames(X)
rownames(Z)<-rownames(X)
gov<-rownames(X)
ds <- dist(Z, method="euclidean")
round(ds, 3)
sinle=hclust(ds, method="single")
plot(sinle, hang=-1, main="Single Linkage
      with euclidean")
ward=hclust(ds, method="ward")
plot(ward, hang=-1, main=" Ward Linkage
      with euclidean")
```

[그림 5.2.]

```
library(biotools)
dm<-D2.dist(X, S)
round(sqrt(dm), 3)
sinlem=hclust(dm, method="single")
plot(sinlem, hang=-1, main="Single Linkage
      with mahalanobis Distance")
wardm=hclust(ds, method="ward")
# wardm=hclust(ds, method="ward.D")
plot(wardm, labels=gov,
      main="Ward Linkage with Mahalanobis Distance")
```

[그림 5.3.]

```
library(NbClust)
Z<-scale(X)
ccc<-NbClust(Z, distance="euclidean", min.nc = 2, max.nc = 8,
             method = "kmeans", index = "ccc")
ccc
plot(2:8, type="b", ccc$All.index, xlab="Number of Clusters",
     ylab="CCC")
dindex<-NbClust(Z, distance="euclidean", min.nc = 2, max.nc = 8,
               method = "kmeans", index = "dindex")
dindex
```

[표 5.2.]

```
allindex<-NbClust(Z, distance="euclidean", min.nc = 2, max.nc = 8,
                 method = "kmeans", index = "all", )
allindex
```

[표 5.3.]

```
Z<-scale(X)
ds <- dist(Z, method="euclidean")
round(ds, 3)
people=rownames(X)
kmeans <- kmeans(Z, 7) # 4 cluster solution
cluster=data.frame(people,cluster=kmeans$cluster)
C1=cluster[(cluster[,2]==1),]
C2=cluster[(cluster[,2]==2),]
C3=cluster[(cluster[,2]==3),]
C4=cluster[(cluster[,2]==4),]
C5=cluster[(cluster[,2]==5),]
C6=cluster[(cluster[,2]==6),]
C7=cluster[(cluster[,2]==7),]
C1;C2;C3;C4;C5;C6;C7
```

[표 5.4.]

```
aggregate(X, by=list(kmeans$cluster),FUN=mean)
```


B. 동학개미운동을 통한 지역/연령별 투자 전략을 이해하기 위한 다변량 자료 분석

1. Data

[작업경로 설정 및 데이터 셋팅]

```
setwd("C:/Users/sec/Desktop/data")  
X = read.csv("stock.csv",header=T)  
rownames(X)<-c( 201:216, 301:316, 401:416,  
               501:516)
```

[분석을 위한 기초적 데이터 변환]

```
X=log(X+1)  
class(X)  
X<-as.matrix(X)  
n<-nrow(X)  
xbar<-t(X)%*%matrix(1,n,1)/n  
I<-diag(n)  
J<-matrix(1,n,n)  
H<-I-1/n*J  
Y<-H%*%X  
S<-t(Y)%*%Y/(n-1)  
D<-diag(1/sqrt(diag(S)))  
Z<-H%*%X%*%D  
colnames(Z)<-colnames(X)  
R<-t(Z)%*%Z/(n-1)
```

[표 1.1.]

```
summary(X)
```

2. Data Description

[그림 2.1.]

```
plot(X)
```

[표 2.1.]

```
cor(X)
```

[표 2.2., 그림 2.2.]

```
#install.packages("MVN")
```

```
#library("MVN") # for mardia test
```

```
result = mvn(X,
```

```
          mvnTest = "mardia",
```

```
          multivariatePlot = "qq")
```

```
result
```

3. Principal Component Analysis(PCA)

[표 3.1.]

```
cov(X)
```

[표 3.2.]

```
eigen.R=eigen(R)
round(eigen.R$values,2)
V=round(eigen.R$vectors,2)
gof=eigen.R$values/sum(eigen.R$values)*100 # Goodness-of fit
round(gof, 2)
```

[그림 3.1.]

```
plot(eigen.R$values, type="b",
      main="Scree Graph",
      xlab="Component Number",
      ylab="Eigenvalue")
```

[표 3.3.]

```
V3=V[,1:3]
V3
P=Z%*%V3
```

[그림 3.2.]

```
plot(P[,1], P[, 2],
      main="Plot of PCs Scores", xlab="1st PC", ylab="2nd PC")
text(P[,1], P[, 2], labels=rownames(X), cex=0.8, col="blue", pos=3)
abline(v=0, h=0)
```

[그림 3.3.]

```
plot(P[,1], P[, 3],
      main="Plot of PCs Scores", xlab="1st PC", ylab="3rd PC")
text(P[,1], P[, 3], labels=rownames(X), cex=0.8, col="blue", pos=3)
abline(v=0, h=0)
```

[그림 3.4.]

```
plot(P[,2], P[, 3],
```

```

    main="Plot of PCs Scores", xlab="2nd PC", ylab="3rd PC")
text(P[,2], P[, 3], labels=rownames(X), cex=0.8, col="blue", pos=3)
abline(v=0, h=0)

```

[그림 3.5.]

```

pcasvd.Z<-prcomp(X, scale=T)
summary(pcasvd.Z)
round(pcasvd.Z$rotation, 3)
pcasvd.Z$scale
biplot(pcasvd.Z, scale=0, xlab="1st PC",ylab="2nd PC",
       main="PC Biplot")
abline(v=0, h=0)

```

4. Factor Analysis (FA)

[FA를 위한 기본 설정]

```

p=ncol(X)
eigen.R=eigen(R)
round(eigen.R$values, 2) # Eigenvalues
V=round(eigen.R$vectors, 2) # Eigenvectors
V

```

[표4.1.]

```

gof=eigen.R$values/p*100 # Goodness-of fit
round(gof, 3) # contribution rate

```

[그림 4.1.]

```

plot(eigen.R$values, type="b", main="Scree Graph",
     xlab="Factor Number", ylab="Eigenvalue")

```

[표 4.3 ,그림 4.2.]

```

install.packages("MVT")
library("MVT")
library(psych)
pcfa<-principal(Z, nfactors=3, rotate="none")
pcfa

```

```

lim<-range(pretty(L))
plot(L[,1], L[,2],main="PC Factor Loadings : f1 and f2", xlab="f1", ylab="f2",
      xlim=lim, ylim=lim)
text(L[,1], L[, 2], labels=rownames(L), cex=0.8, col="blue", pos=1)
abline(v=0, h=0)
arrows(0,0, L[,1], L[, 2], col=2, code=2, length=0.1)
plot(L[,1], L[,3],main="PC Factor Loadings : f1 and f3", xlab="f1", ylab="f3",
      xlim=lim, ylim=lim)
text(L[,1], L[, 3], labels=rownames(L), cex=0.8, col="blue", pos=1)
abline(v=0, h=0)
arrows(0,0, L[,1], L[, 3], col=2, code=2, length=0.1)
plot(L[,2], L[,3],main="PC Factor Loadings : f2 and f3", xlab="f2", ylab="f3",
      xlim=lim, ylim=lim)
text(L[,2], L[, 3], labels=rownames(L), cex=0.8, col="blue", pos=1)
abline(v=0, h=0)
arrows(0,0, L[,2], L[, 3], col=2, code=2, length=0.1)

```

[그림 4.3.]

```

fpc=pcfa$scores
round(fpc, 3)
par(pty="s")
rownames(fpc)<-c( 201:216, 301:316, 401:416,
                 501:516)
lim<-range(pretty(fpc))
plot(fpc[,1], fpc[,2],main="Factor Scores : f1 and f2", xlab="f1", ylab="f2",
      xlim=lim, ylim=lim)
text(fpc[,1], fpc[,2], labels=rownames(fpc), cex=0.6, col="blue")
abline(v=0, h=0)
plot(fpc[,1], fpc[,3],main="Factor Scores : f1 and f3", xlab="f1", ylab="f3",
      xlim=lim, ylim=lim)
text(fpc[,1], fpc[,3], labels=rownames(fpc), cex=0.6, col="blue")
abline(v=0, h=0)
plot(fpc[,2], fpc[,3],main="Factor Scores : f2 and f3", xlab="f2", ylab="f3",
      xlim=lim, ylim=lim)
text(fpc[,2], fpc[,3], labels=rownames(fpc), cex=0.6, col="blue")
abline(v=0, h=0)

```

[그림 4.4.]

```
svd.Z <- svd(Z)
U <- svd.Z$u
V <- svd.Z$v
D <- diag(svd.Z$d)
F <- (sqrt(n-1)*U)[,1:2] # Factor Scores Matrix : F
L <- (sqrt(1/(n-1))*V%*%D)[,1:2] # Factor Loadings Matrix : Lambda
C<- rbind(F, L)
rownames(F)<-rownames(X)
rownames(L)<-colnames(X)
varimax<-varimax(L)
Lt = varimax$loadings
T=varimax$rotmat
T
Ft= F%*%T
biplot(Ft,Lt, xlab="f1",ylab="f2", main="Varimax Rotated Biplot",
       xlim=c(-4,4),ylim=c(-4.2,2),cex=0.5,pch=16)
abline(v=0,h=0)
```

[표 4.4]

```
L=pcfa$loading[, 1:3]
round(L, 3)
Psi=pcfa$uniquenesses
Psi
Rm = R-(L%*%t(L) + diag(Psi))
round(Rm, 3)
```

5. Cluster Analysis (CA)

[그림 5.1.]

```
n<-nrow(X)
xbar<-t(X)%*%matrix(1,n,1)/n
I<-diag(n)
J<-matrix(1,n,n)
H<-I-1/n*J
Y<-H%*%as.matrix(X)
S<-t(Y)%*%Y/(n-1)
D<-diag(1/sqrt(diag(S)))
Z<-Y%*%D
colnames(Z)<-colnames(X)
rownames(Z)<-rownames(X)
gov<-rownames(X)
ds <- dist(Z, method="euclidean")
round(ds, 3)
sinle=hclust(ds, method="single")
plot(sinle, hang=-1, main="Single Linkage
      with euclidean")
ward=hclust(ds, method="ward")
plot(ward, hang=-1, main=" Ward Linkage
      with euclidean")
```

[그림 5.2.]

```
install.packages("biotools")
library(biotools)
dm<-D2.dist(X, S)
round(sqrt(dm), 3)
sinlem=hclust(dm, method="single")
plot(sinlem, hang=-1, main="Single Linkage
      with mahalanobis Distance")
wardm=hclust(ds, method="ward")
# wardm=hclust(ds, method="ward.D")
plot(wardm, labels=gov,
      main="Ward Linkage with Mahalanobis Distance")
```

[그림 5.3.]

```
install.packages("NbClust")
library(NbClust)
Z<-scale(X)
ccc<-NbClust(Z, distance="euclidean", min.nc = 2, max.nc = 8,
             method = "kmeans", index = "ccc")
ccc
plot(2:8, type="b", ccc$All.index, xlab="Number of Clusters",
     ylab="CCC")
dindex<-NbClust(Z, distance="euclidean", min.nc = 2, max.nc = 8,
               method = "kmeans", index = "dindex")
index
```

[표 5.1.]

```
allindex<-NbClust(Z, distance="euclidean", min.nc = 2, max.nc = 8,
                 method = "kmeans", index = "all", )
allindex
```

[표 5.2.]

```
Z<-scale(X)
ds <- dist(Z, method="euclidean")
round(ds, 3)
people=rownames(X)
kmeans <- kmeans(Z, 8) # 4 cluster solution
cluster=data.frame(people,cluster=kmeans$cluster)
C1=cluster[(cluster[,2]==1),]
C2=cluster[(cluster[,2]==2),]
C3=cluster[(cluster[,2]==3),]
C4=cluster[(cluster[,2]==4),]
C5=cluster[(cluster[,2]==5),]
C6=cluster[(cluster[,2]==6),]
C7=cluster[(cluster[,2]==7),]
C8=cluster[(cluster[,2]==8),]
C1:C2:C3:C4:C5:C6:C7:C8
```

[표 5.3.]

```
aggregate(X, by=list(kmeans$cluster),FUN=mean)
```


[丑 5.4.]

```
aggregate(10^(X+1), by=list(kmeans$cluster),FUN=mean)
```

C. 결론

[데이터 분석을 위한 기초 작업]

```
setwd("C:/Users/sec/Desktop")
X = read.csv("final.csv",header=T)
rownames(X)<-c( 201:216, 301:316, 401:416,
               501:516)
n<-nrow(X)
xbar<-t(X)%*%matrix(1,n,1)/n
I<-diag(n)
J<-matrix(1,n,n)
H<-I-1/n*J
Y<-H%*%as.matrix(X)
S<-t(Y)%*%Y/(n-1)
D<-diag(1/sqrt(diag(S)))
Z<-Y%*%D
colnames(Z)<-colnames(X)
rownames(Z)<-rownames(X)
gov<-rownames(X)
ds <- dist(Z, method="euclidean")
```

[그림 C.1.]

```
#install.packages("NbClust")
library(NbClust)
Z<-scale(X)
ccc<-NbClust(Z, distance="euclidean", min.nc = 2, max.nc = 8,
             method = "kmeans", index = "ccc")
ccc

allindex<-NbClust(Z, distance="euclidean", min.nc = 2, max.nc = 8,
                 method = "kmeans", index = "all", )
allindex
```

<C.표1.두 데이터를 통합했을 때의 군집 분할>

```
Z<-scale(X)
ds <- dist(Z, method="euclidean")
people=rownames(X)
```

```
kmeans <- kmeans(Z, 6) # 4 cluster solution
cluster=data.frame(people,cluster=kmeans$cluster)
C1=cluster[(cluster[,2]==1),]
C2=cluster[(cluster[,2]==2),]
C3=cluster[(cluster[,2]==3),]
C4=cluster[(cluster[,2]==4),]
C5=cluster[(cluster[,2]==5),]
C6=cluster[(cluster[,2]==6),]
C1:C2:C3:C4:C5:C6:
```